

Solution Code - Microservices - Resilience4j_Circuit_Breaker

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\example\microservices\CircuitBreakerApplication.java

```
1 | package com.example.microservices;
2 |
3 | import org.springframework.boot.SpringApplication;
4 | import org.springframework.boot.autoconfigure.SpringBootApplication;
5 | import org.springframework.web.bind.annotation.*;
6 | import io.github.resilience4j.circuitbreaker.annotation.CircuitBreaker;
7 |
8 | @RestController
9 | class DemoController {
10 |     @GetMapping("/data")
11 |     @CircuitBreaker(name = "backendService", fallbackMethod = "fallbackData")
12 |     public String getData() {
13 |         throw new RuntimeException("Service failure simulation");
14 |     }
15 |
16 |     public String fallbackData(Throwable t) {
17 |         return "Resilience4j Fallback Data Response";
18 |     }
19 | }
20 |
21 | @SpringBootApplication
22 | public class CircuitBreakerApplication {
23 |     public static void main(String[] args) {
24 |         SpringApplication.run(CircuitBreakerApplication.class, args);
25 |     }
26 | }
```

Source File: src\main\java\com\example\microservices\ServiceApplication.java

```
1 | package com.example.microservices;
2 |
3 | import org.springframework.boot.SpringApplication;
4 | import org.springframework.boot.autoconfigure.SpringBootApplication;
5 | import org.springframework.cloud.client.discovery.EnableDiscoveryClient;
6 |
7 | @EnableDiscoveryClient
8 | @SpringBootApplication
9 | public class ServiceApplication {
10 |     public static void main(String[] args) {
11 |         SpringApplication.run(ServiceApplication.class, args);
12 |         System.out.println("Microservice Exercise 4 running...");
13 |     }
14 | }
```